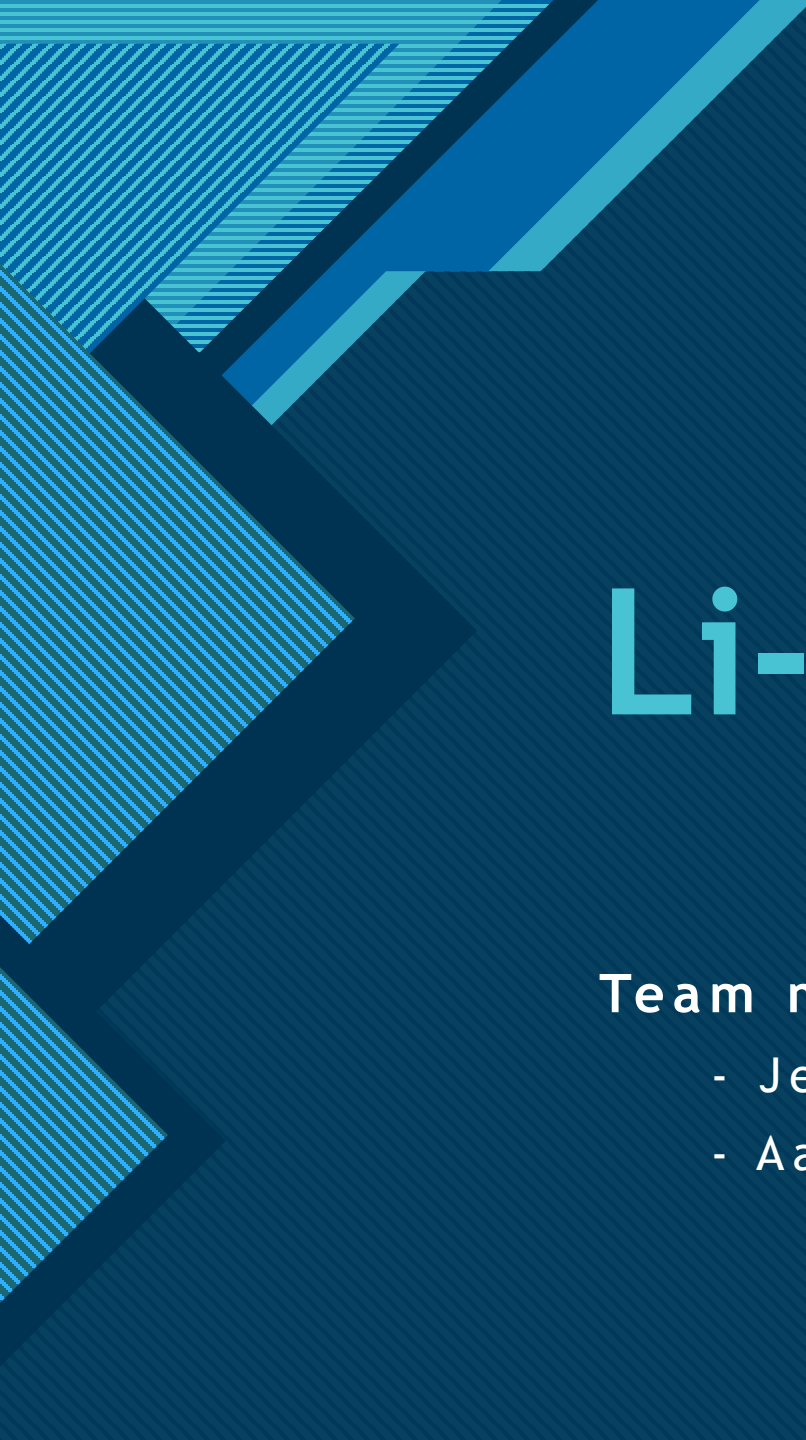




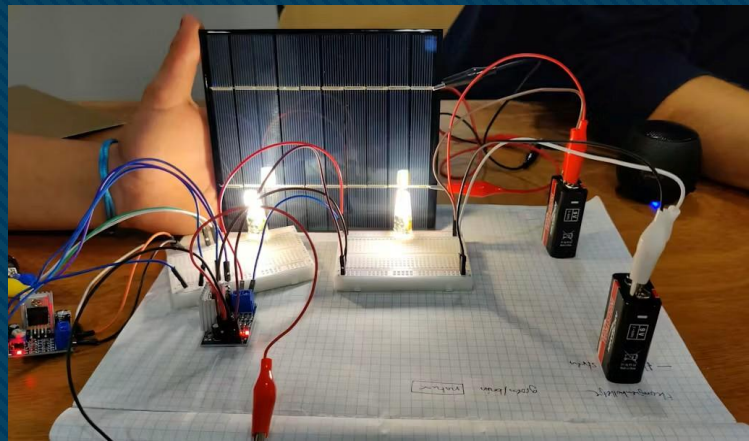
Li-Fi Based Data Transmission

Team members :

- Jeel Sidpara
- Aatman Sabhaya

Li-Fi Based Data Transmission

- Li-Fi - Light Fidelity (It is a wireless communication technology that uses light)
- This project demonstrates wireless transmission of data signals using visible light communication (Li-Fi) by modulating LED intensity and detecting it with a photodiode



Agenda :

- Problem
- Solution
- Working
- Resource
- Use Cases

Li-Fi reduces existing problems :

Examples

- **Wi-Fi spectrum is congested:**
Too many devices cause interference and network overload.
- **Limited speed in high-density areas:**
Shared bandwidth reduces speed for users.
- **Security issues (signals pass through walls):**
Signals can be accessed outside, risking unauthorized access.
- **Not suitable in RF-sensitive environments:**
Radio signals can interfere with critical equipment.

Solution :

- **Uses visible light (LED) for communication:**

Data is transmitted by rapidly varying the intensity of LED light, which is not visible to the human eye.

- **No radio frequency usage:**

It operates using light waves instead of RF, avoiding interference with wireless networks and sensitive devices.

- **High-speed & secure transmission:**

Light allows very high data rates and cannot pass through walls, ensuring faster and more secure communication.

Working :

The data signal is converted into an electrical signal and used to modulate the LED's light intensity.

These light variations transmit data through air. A solar panel or photodiode receives the light, converts it back into an electrical signal, which is amplified and transmitted through a output device

Data → LED → Light beam → Solar panel/Photodiode → Output Device



Resource :

- LED
- Solar Panel / Photodiode
- Jump Wires
- Battery
- Resistors
- Microcontroller – Arduino Uno
- Speaker / output device
- Sensor

Use Cases :

1. Hospitals & Medical Environments:

- Eliminates RF interference with sensitive equipment like MRI machines and pacemakers
- Enables secure, localized patient data transmission within wards and ICUs

2. Aircraft Cabin Connectivity:

- Delivers in-flight internet to passengers via overhead LED lights without affecting avionics
- Provides interference-free, high-speed data access in RF-restricted flight zones

3. Industrial Automation & Factory Floors:

- Offers stable communication for sensors and robots in high electromagnetic interference environments
- Supports real-time machine-to-machine data exchange in Industry 4.0 smart factories



Thank You